

SENIOR SOFTWARE / AI ENGINEER

Roman Girin

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Professional Summary

Senior Software & AI Engineer with 20+ years of experience specializing in high-load systems, enterprise solution architecture modernization, cloud migrations, AI integration (LLMs, CNNs) and ML. Combines a strong academic foundation (PhD in Neural Networks) with a proven track record of building commercial systems.

Core Competencies & Tech Stack

Programming Languages: C#, Python, C++, SQL (T-SQL)

Architecture: Monolith-to-Microservices, 2-tier-to-3-tier, OpenAPI, EDA, CNCF, OpenTelemetry API, CQRS, Saga, DDD

Software Engineering & Frameworks: .NET 8 / Core / Framework, ASP.NET, Django / Django REST

AI Agentic Workflows / Machine Learning: CNN, LLM, PyTorch, OpenCV, TorchVision, Transformers, RAG, pgvector, LoRA Fine-Tuning, MS Semantic Kernel

AI Tools: Claude Code, LM Studio

Data Science & Analytics: NumPy, Pandas, Scikit-Learn, Imbalanced-Learn (SMOTE)

Databases: MS SQL, PostgreSQL, noSQL Azure Table Storage

Cloud & Microservices: Docker, Azure Kubernetes Service (AKS) / Container Apps / Instances, Azure Event Grid / Queues, CI/CD, Azure DevOps (CI/CD), Kafka, RabbitMQ.

PROFESSIONAL EXPERIENCE

RiverLogic Inc. | Seattle, USA (Remote)

Senior developer, Modelling software Value Chain Optimization (VCO) | March 2014 – Present

- **Architectural Evolution:** Spearheaded the modernization from legacy 2-tier to 3-tier architecture with zero downtime during migration. Migrated Business Logic Layer (BLL) from MSSQL T-SQL to .NET C# application server, improving system maintainability resulting in a 30% reduction in support ticket volume. Implemented compatibility with PostgreSQL.
- **Windows to Linux Migration:** Migrated the core codebase from .NET Framework to .NET Core / .NET 8 to achieve full cross-platform compatibility, introduction await and optimize cloud performance resulting in a 15% reduction in system resources consumption.
- **Cloud Deployment:** Decomposed an on-premise monolith into high-load microservices deployed on Azure Container Apps and Azure Kubernetes Service (AKS). Designed inter-

service communication using REST and asynchronous messaging EDA (Azure Event Grid / Queues), ensuring distributed data consistency via Saga, CQRS, and Outbox patterns.

- **AI Assistant Development:** Implemented RAG, embedding vector search. Built an MS Semantic Kernel plugin to integrate core software via ASP.NET 8 REST HATEOAS API.
- **Python-Driven LLM Fine-Tuning:** Conducted high-impact R&D and executed LLM fine-tuning via LoRA using the Transformers library (Python, PyTorch) to customize, optimize, and align open-source models for domain-specific enterprise workflows.
- **Interoperability:** Developed critical components in C++ to seamlessly integrate CLR-managed and native code.
- **Quality Assurance and Refactoring:** Practice DDD, SOLID, GoF. Established testing standards by implementing comprehensive unit and integration testing (xUnit, NUnit) and Test-Driven Development (TDD).
- **Team Upskilling:** Upskilled the engineering team on agentic AI workflows using Claude Code, decreasing repetitive code drafting and increasing overall development velocity. Managed build and deployment workflows via Azure DevOps pipelines.

Samara State Technical University | Samara, Russia

Machine Vision Researcher & Developer (Part-Time, Research Fellowship) | September 2017 – September 2019

- **Deep Learning:** Engineered a machine vision diagnostic system utilizing Deep Convolutional Neural Networks (CNNs) to automate the detection of broken railway fasteners and bindings using Python, PyTorch, OpenCV, TorchVision, and CUDA/cuDNN acceleration.
- **Data Engineering & Optimization:** Managed, analyzed, and prepared large-scale imagery datasets using NumPy, Pandas, and Scikit-Learn for training, validation, and test splits to ensure robust model generalizability.
- **Imbalanced Data Handling:** Addressed severe class imbalances within the training data by implementing data augmentation techniques, including SMOTE (Synthetic Minority Over-sampling Technique) via the imbalanced-learn library.
- **Integration & Deployment:** Implemented cross-system integration via REST HATEOAS API (Django REST) and asynchronous messaging (Kafka, RabbitMQ). Successfully deployed the diagnostic system in a production environment for JSC RPC InfoTrans increasing average number of registered railway defects by 60%.

VDcom | Samara, Russia

Senior Specialist (Promoted for outstanding performance) | October 2011 – March 2014

Specialist | May 2010 – October 2011

- **Outstaffing at Sberbank CIB:** Developed backend functionality on the Diasoft platform for the investment banking business.
- **High-load Databases:** Designed, developed, and optimized complex stored procedures, ER-models and data-tier components for high-load transaction systems across Sybase ASE and MSSQL database engines.

- **Technical Estimation:** Conducted technical analysis and pre-sale engineering evaluations for resource and cost estimation.

ACS – VolgaEVMcomplex | Samara, Russia

Project Manager / System Analyst | *February 2006 – May 2010*

- **Requirements Engineering:** Formalized client requirements, authoring comprehensive technical specifications (SRS) to automate business processes on the DocsVision platform.
- **Management & .NET Development:** Managed EDMS (Electronic Document Management Systems) implementation projects. Led a full-stack development team while directly contributing to the development of custom extensions and C# integration modules.

EDUCATION & RESEARCH

The Samara State Technical University | Samara, Russia

PhD in Technical Sciences (Artificial Neural Networks Focus) | *Graduated 2019*

- **Thesis:** “Intelligent informational measuring system for thermography-based diagnostic of technical objects based on artificial neural networks.”
- **Research Portfolio:** Published academic insights available at ResearchGate <https://www.researchgate.net/profile/roman-qirin/research> .

The Samara State Technical University | Samara, Russia

Specialist Degree in Engineering | *Graduated 2002*

CREDENTIALS & SKILLS

- **Certifications:** Federal Financial Markets Service (FFMS) Professional Certificate (2008).
- **Languages:** Russian (Native), English (Full Professional Proficiency).
- Designed, implemented, and deployed into production a series of pet projects; descriptions and demos are available upon request.